



Rootkit

What is a rootkit?

The evil rootkit is a set of software tools usually used by an intruder after gaining access to a computer. The tools are intended to conceal running processes, files or system data, which helps the intruder maintain access to the system without the user's knowledge.

How does it work?

A rootkit integrates itself deep within the OS, taking over bits and pieces of the code, and then hiding from anything that is trying to detect it. Consider a regular OS file. The rootkit might infect the file and become part of it. If you try and use an anti-virus or anti-spyware against it, the rootkit performs a trick that in effect tells the anti- software that there's nothing wrong with the file. Thus, the rootkit is completely invisible.

Who developed it?

"Rootkit" originally referred to a set of recompiled Unix tools that would hide any trace of an intruder; the tools would normally display such traces. This allowed the intruder to be a "root" on the system without being visible to the administrator. The term is now not restricted to Unix-based OSes; such tools exist for OSes such as Windows.

Where is it used?

Usually on compromised computers, and famously, on Sony music CDs until recently.

When did the term become popular?

In October 2005, a certain blog detailed a description of the characteristics of the software on Sony BMG music CDs. The article said the software was illegitimate. It was stated that there were shortcomings in the software design that manifest themselves as security holes, which can be exploited by malicious software.

Why are rootkits legitimately needed?

Rootkits allow the programmer to see and access usernames and login information for sites that require them.

through ads embedded within recorded programmes, and, yes, even channel-surfing during ad breaks. Viewers will be forced to pay if they want to avoid the ads.

"A viewer may either watch the ads or pay a fee in order to be able to change channels or fast-forward when the ads are being displayed," states Philips' would-be patent.

Philips' application for the patent was originally filed in 2003, but was only recently made available online. It outlines how digital flags would be transmitted with the ads—a TV, PVR, DVD recorder or other receiver would detect the flag and ensure the ads are shown no matter what. The exact details of how this works aren't clear.

Why has a 2003 patent not made it to market yet? Philips admits the idea might be "greatly resented by the viewers," and suggests that broadcasters and device makers present warnings when the flag is detected and used. They acknowledge that the "manufacturer of a video display device (such as a set top box) has no way to ensure that a program broadcaster will notify the viewers and explain why the viewers are no longer able to switch channels during advertisements." Yes, it's a difficult proposition indeed.

However, "resented" as it may be by viewers—that includes you—we think something like this is bound to make it way to market some time or the other, whether we're talking about the Web or television. For example, ad-free HBO in the US costs a pretty penny each month, while in India, it's virtually ad-free because we simply switch channels—and we pay a pittance for the channel. How long can such a situation last? And on the Web, it's much easier to force you to watch ads—there isn't even a new technology required! How

long before all sites require you to watch ads before content is displayed?

There has been a trend on the Net of forcing people to watch ads. Some years ago, there were hardly any sites that required you to register in order to view their content—and with your mail ID, they now send you ads by mail. Then there's Salon.com: you need to get a "day pass" to get to the site, and you do that by watching an ad for a while. Then there are these gentler sites that say, "Such-and-such is brought to you by our sponsor—click here to continue." And the sponsor's ad appears below that text, but who bothers to wait for it to load? You just click the button!

We can't predict *when* this will change, and only time will tell—but it's bound to happen.

AN ANTI-PIRACY CHINA?

Computers Must Have OSes

Wang Yefei, a senior executive from the Beijing Copyright Bureau, recently announced that China would crack down on system assemblers who sell systems without operating systems, as this was leading to piracy.

He also announced that no government department would buy computers without OSes, and all locally-made or imported computers would require to be shipped with the OS.

Obviously, reporters were quick to ask if the government had thus far been purchasing computers without operating systems, or loading pirated software... the response was a hasty statement that all government agencies use legal software!

Jokes apart, the move on the Chinese government's part might actually help reduce piracy in China, which has been cracking down on software pirates since early 2000. Statistics state that China has "detained" over 13,000 Intellectual Property Rights (IPR) violators, who have caused losses of an estimated \$600 million (Rs 2,700 crore).

Of course, China can afford to police its people the way it does, while other countries cannot. Imagine if you were being forced to buy a computer along with Windows XP just because your system assembler didn't know how to install your favourite Linux distro: sure it would help curb piracy a little, but do we really not care about all the software, other than the OS, that is pirated?

Whether or not this move actually helps China crack down on IPR violators only time will tell. In the meanwhile, it's something for our own Government to mull over. ■

